Power



File: power.kicad_sch

Power distribution: see sheet 'Power' (power-analysis.md, 2026-10-27).
Control / signal sheet to follow.

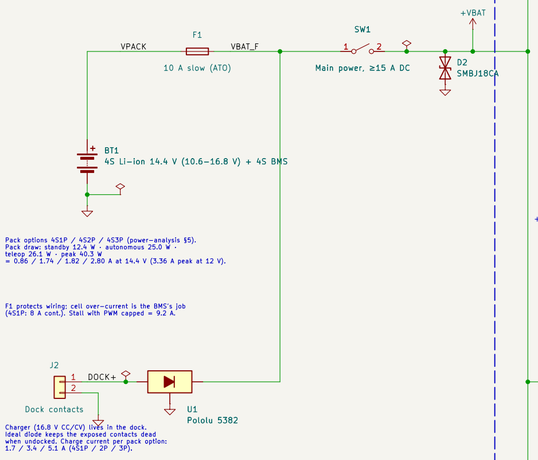
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Sheet: /
File: RoamView.kicad_sch

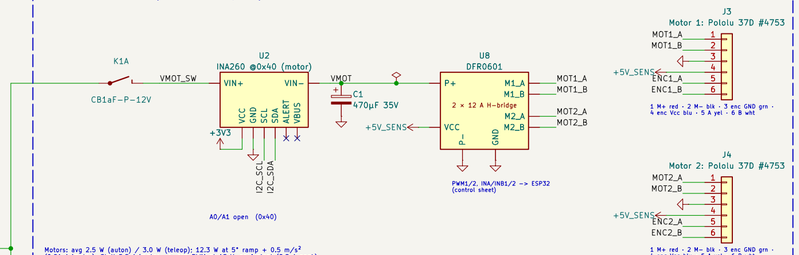
Title: RoamView

Size: A4	Date: 2026-10-27	**Rev: A (draft)**
KiCad E.D.A. 10.0.6		Id: 1/2

1. PACK, CHARGE INPUT & MAIN PROTECTION

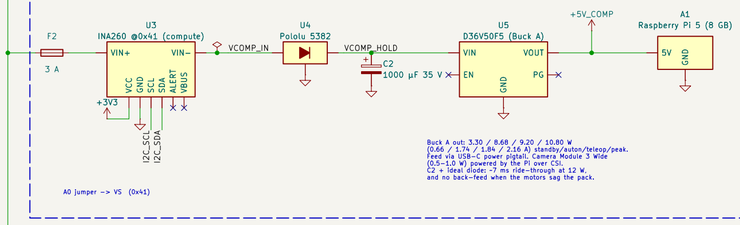


2. MOTOR BRANCH (direct from pack)

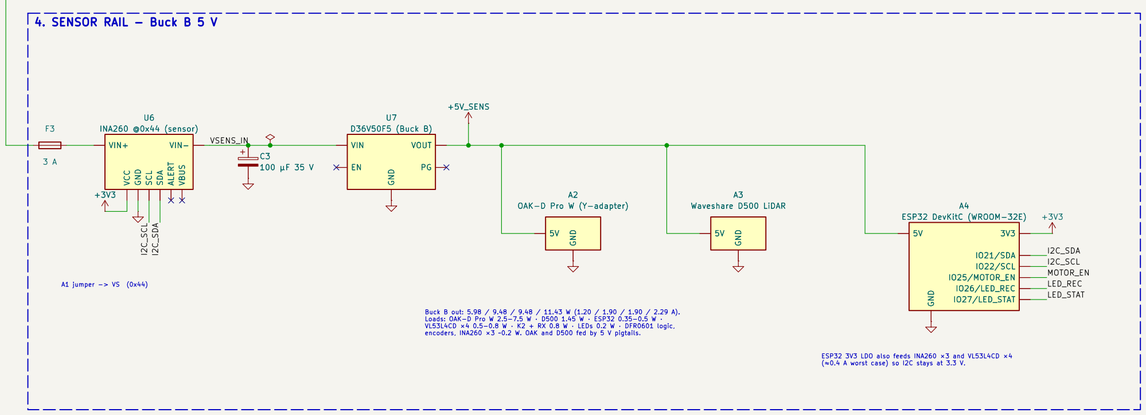


+VBAT bus

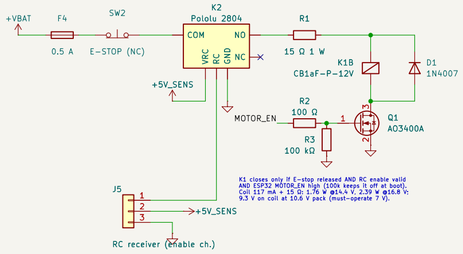
3. COMPUTE RAIL — Buck A 5 V



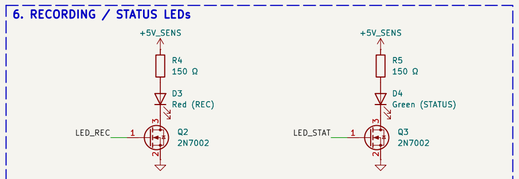
4. SENSOR RAIL — Buck B 5 V



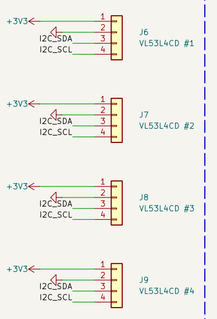
5. MOTOR ENABLE / E-STOP (K1 coil)



6. RECORDING / STATUS LEDs



7. CLIFF SENSORS (3V3 I2C)



NOTES

Source: hardware/power-analysis.md (2026-10-27) and Power Arch. 2026-10-27.drawio.png. Module-level wiring schematic:
Pololu, Adafruit, DFRobot and Luxonis parts are off-the-shelf boards; footprints are left blank on purpose.

RAILS
- VPACK — pack + (BT1) to F1 — 10.6–16.8 V
- VBAT_F — after F1: charge diode U1 lands here so charging works with SW1 off
- +VBAT — switched bus after SW1, D2 TVS clamps spikes (SMBJ18CA, 29 V clamp < 36 V INA260 limit)
- VMOT_SW / VMOT — motor branch after K1 and after U2 shunt: C1 bulk at DFR0601 P+
- +5V_COMP — Buck A (U5) -> Raspberry Pi 5 only. U4 ideal diode + C2 isolate it from motor sag
- +5V_SENS — Buck B (U7) -> OAK, D500, ESP32, DFR0601 logic, encoders, K2, RC RX, LEDs
- +3V3 — ESP32 board LDO -> INA260 ×3 VCC, VL53L4CD >4 VIN (keeps I2C at 3.3 V for the ESP32)

CURRENT SENSE (INA260, 2 mΩ internal, high side)
- U2 0x40 motor (A0/A1 open) · U3 0x41 compute (A0 -> VS) · U6 0x44 sensor (A1 -> VS), I2C master: ESP32 IO21/IO22.
- INA260 VCC = 3V3, so the breakout pull-ups go to 3.3 V. ALERT unused; VBUS jumpered to VIN− on the breakout.

FUSING
- F1 10 A main (ATO slow): wiring short-circuit protection. Normal peak 2.8 A ⌀14.4 V: capped stall 9.2 A.
- F2 / F3 3 A per buck branch: buck input <= 1.2 A at 10.6 V and 11–12 W out, plus C2/C3 inrush.
- F4 0.5 A on the K1 coil chain (142 mA max at 16.8 V).

MOTOR ENABLE (spec: power-analysis §2.3 #14, firmware e-stop interlock)
- K1 coil = +VBAT -> F4 -> SW2 E-stop (NC) -> K2 RC relay (NO) -> R1 15 Ω -> coil -> Q1 (ESP32 MOTOR_EN).
- Any one of the three opens K1 and removes motor power; RC loss de-energises K2 (fail-safe).

OPEN ITEMS / CHECK BEFORE BUILD
1. D36V50F5 is 5.0 V ±4 %; the Pi 5 wants 5.1 V. Check for under-voltage warnings under load; add usb_max_current_enable=1 only if the PI USB load stays well under the buck's budget.
2. Pack chemistry/size not final: re-check F1 against the chosen pack's BMS limit.
3. Confirm #2804 relay fail-safe behaviour on RC signal loss on the bench.
4. Validate every figure with the three INA260 readings (power-analysis header).
5. Control signals (DFR0601 PWM/IN, encoder A/B, VL53 XSHUT, LIDAR UART) belong on a control sheet.

Module-level power wiring; values to confirm against INA260 measurements
Derived from hardware/power-analysis.md (2026-10-27)

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